

# 2010376103 CSE4261 Assignment-11

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July 2025

## 1 Comparing three face verifiers which are trained using Binary Cross-Entropy loss, contrastive loss and triplet loss, receptively.

[Github code link](#)

### 1.1 Dataset

The experiments were conducted on the **Labeled Faces in the Wild (LFW)** dataset, a benchmark for face verification tasks. The dataset consists of face images of celebrities collected from the web. For training and evaluation:

- The data pairs were generated using the standard LFW pairs.txt protocol.
- Same-identity and different-identity pairs were extracted from this list.
- Images were resized and converted to RGB format.
- All samples were loaded using a custom PyTorch dataset loader, LFWPairDataset, with optional data transformations.

### 1.2 Model

#### Embedding Network:

A shared convolutional neural network is used for all three losses. It includes:

- Three convolutional layers with ReLU activations.
- Max-pooling and adaptive average pooling to reduce spatial dimensions.
- A final fully connected layer projecting the image into a 128-dimensional embedding space.

#### Loss Function Variants:

- **Binary Cross-Entropy (BCE):** A Siamese network with cosine similarity followed by a sigmoid activation and BCE loss.
- **Contrastive Loss:** Embeddings are compared using Euclidean distance. The loss pulls similar pairs together and pushes dissimilar ones apart beyond a fixed margin.
- **Triplet Loss:** Uses triplets (anchor, positive, negative) to train the model to bring positive pairs closer and separate negatives in the embedding space.

#### Training Configuration:

- Optimizer: Adam
- Learning Rate: 0.0001
- Batch Size: 128
- Number of Epochs: 15
- Hardware: GPU-enabled PyTorch environment

### 1.3 Result and Discussion

The three models were evaluated based on verification accuracy and visual similarity for selected pairs from the test set.

## Accuracy Comparison

Table 1 shows the performance of the three face verification models trained with BCE, Contrastive, and Triplet loss functions. It includes overall accuracy, predicted similarity score for a representative pair, and the model’s final decision.

Table 1: Comparison of accuracy, similarity, and decision across loss functions

| Model       | Accuracy | Similarity | Decision |
|-------------|----------|------------|----------|
| BCE         | 95.73%   | 0.72       | Same     |
| Contrastive | 93.42%   | 0.91       | Same     |
| Triplet     | 94.19%   | 0.79       | Same     |

The BCE model achieved the highest accuracy, showing that using cosine similarity with BCE is effective when paired samples are well-structured and pre-aligned.

## Qualitative Verification

Figure 1 presents a visual comparison of face verification results using models trained with BCE, Contrastive, and Triplet loss functions. Each column corresponds to one of the loss types, showing paired input images along with the model’s prediction—whether the pair represents the same individual or not.



Figure 1: Comparison among BCE, Contrastive and Triplet

## Loss Behavior

The BCE model converged faster due to the direct binary supervision. The contrastive model showed a more gradual decrease in loss, as it requires the model to learn from both similar and dissimilar distances. The triplet loss training was more sensitive to sample selection but provided meaningful separation in embedding space when trained well. Figure 2 shows the loss curve.

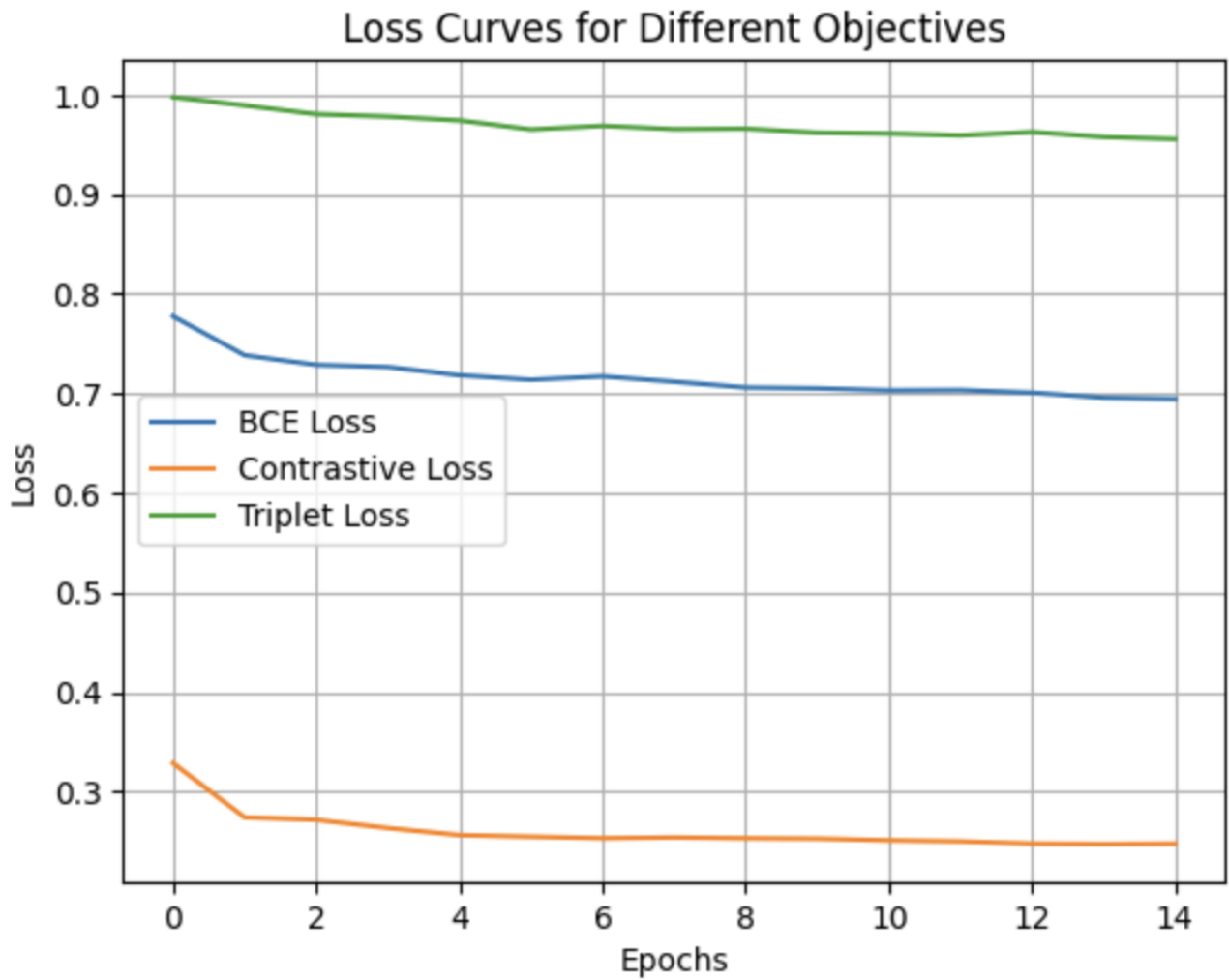


Figure 2: Loss among BCE, Contrastive and Triplet

## Summary

Each loss offers unique strengths:

- BCE excels in binary match classification tasks with simple network outputs.
- Contrastive loss encourages margin-aware distance learning.
- Triplet loss ensures global ranking and embedding separation.

For applications focused on pairwise decision-making, BCE provides high performance. However, contrastive and triplet losses may generalize better in open-set or retrieval settings due to structured embedding learning.

## 2 Comparing effect of Binary Cross-Entropy (BCE) loss and Mean Squared Error (MSE) as a reconstruction loss during training of a Variational Autoencoder (VAE).

[Github code link](#)

### 2.1 Database

The experiment was conducted on the MNIST dataset, which consists of handwritten digits. Key features of the dataset are:

- Number of training samples: 60,000
- Number of testing samples: 10,000
- Image resolution:  $28 \times 28$  pixels
- Channel: Single channel (grayscale)
- Classes: 10 (digits from 0 to 9)

### 2.2 Model

The core model used for this experiment is the **Variational Autoencoder (VAE)**. The VAE consists of two primary components: an encoder and a decoder. The encoder maps the input to a latent space using a Gaussian distribution, while the decoder reconstructs the original input from the latent vector.

#### Network Architecture:

- **Encoder:**
  - Input layer: 784 neurons (flattened 28x28 image)
  - Hidden layer: 400 neurons with ReLU activation
  - Latent mean ( $\mu$ ) and log-variance ( $\log \sigma^2$ ): 20 neurons each
- **Reparameterization:**
  - $z = \mu + \sigma \cdot \epsilon$ , where  $\epsilon \sim \mathcal{N}(0, 1)$
- **Decoder:**
  - Hidden layer: 400 neurons with ReLU activation
  - Output layer: 784 neurons with Sigmoid activation

#### Training Setup:

- Optimizer: Adam
- Learning Rate: 0.001
- Epochs: 5
- Batch Size: 128
- Two separate models were trained:
  - One using **Binary Cross-Entropy (BCE)** loss
  - One using **Mean Squared Error (MSE)** loss

### 2.3 Result and Discussion

#### Pixel-wise Accuracy:

- VAE with BCE Loss: **0.9573**
- VAE with MSE Loss: **0.9548**

Both models achieve high accuracy, but the BCE-trained VAE shows a slightly higher pixel-wise reconstruction accuracy. Since MNIST images are binary in nature (mostly consisting of foreground digits on a black background), BCE is more aligned with the data distribution, treating each pixel as a Bernoulli trial.

### Visual Comparison of Reconstructions:

Figure 3 describes BCE-based reconstructions retain better edge definition and digit clarity. The MSE-based model produces slightly blurrier outputs, which is expected since MSE penalizes large deviations more evenly across all pixel values, leading to smoother reconstructions. On the other hand, BCE loss tends to encourage sharper binary predictions, which is advantageous in datasets like MNIST.

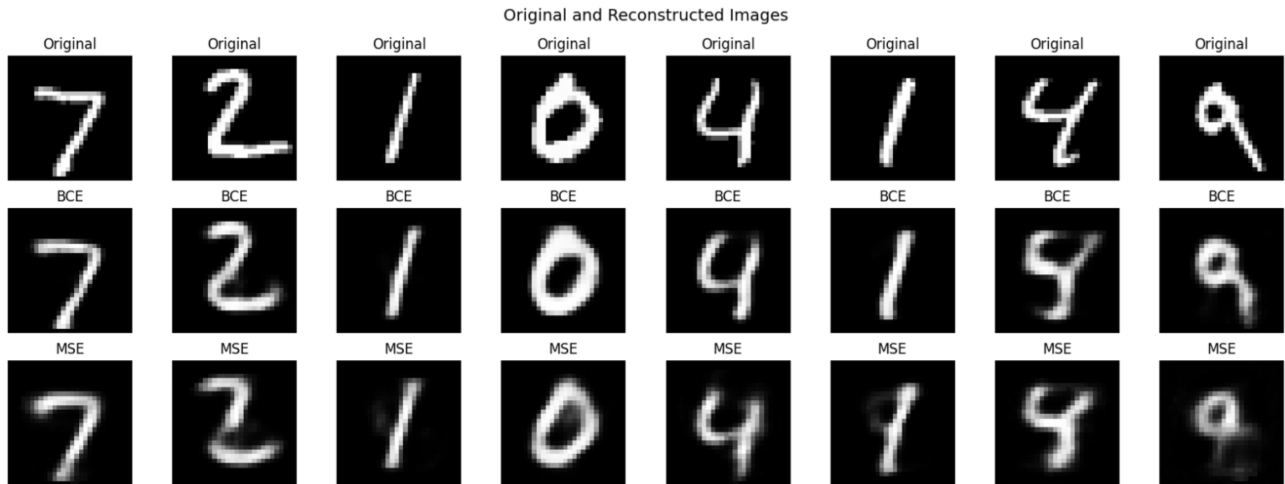


Figure 3: Reconstructed Images by VAE

### Loss Landscape:

During training, BCE loss exhibited faster convergence and smaller reconstruction loss compared to MSE. However, MSE is known to generalize better in natural image domains where pixel intensities are continuous rather than binary.

### Conclusion:

In binary image domains like MNIST, Binary Cross-Entropy provides better pixel-wise accuracy and visually sharper reconstructions. While both losses are valid, BCE is more suitable for binary data, and MSE might be more effective in grayscale or continuous-valued domains.